

Gali Hari Santhosh

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Professional Summary

Enthusiastic Data Analyst with a strong foundation in data analysis, statistical modeling, and machine learning. Proficient in Python, SQL, Tableau, and Power BI. Experienced in creating dashboards and building predictive models. Holds Data Science certifications from Besant Technologies and Reliance Foundation. Skilled in data preprocessing, visualization, and deriving actionable insights from complex datasets. Eager to leverage technical expertise and analytical skills to solve business challenges and drive data-informed decision-making in a dynamic organization.

Education

KL University 2025 – 2027
M.Tech in Electronics and Communication Engineering (ECE), Specialization: VLSI, CGPA: 8.55

New Prince Shri Bhavani College of Engineering and Technology Sept 2018 – May 2022
B.E in Electronics and Communication Engineering, CGPA: 7.9

Viswasai Junior College Mar 2016 – May 2018
Intermediate in MPC, Marks: 931

Sri Sainadh High School June 2016
10th Grade, CGPA: 9.2

Training

Data Science Training Program, Besant Technologies Institute May 2024 – Nov 2024
Bangalore, Kalyan Nagar, India

- Proficiency in Python programming for data science.
- Skills in using SQL for data manipulation and database management.
- Completed a Python project on Banking Application, demonstrating problem-solving and programming skills.
- Leveraged Python programming skills to conduct a Telco Customer Churn prediction project, as part of the data science training program.

Certification

- **Certificate of Appreciation – AI/ML-Powered FPGA Design & Simulation Hackathon** (2025): Awarded Rank-16 in the hackathon organized by NIT Jamshedpur in collaboration with VLSI FOR ALL, as a tribute to Sir Ratan Tata.
- **Data Science Essentials Certificate – Reliance Foundation** (Nov 2024): Covered Python, SQL, data visualization, and machine learning concepts with hands-on projects in data analysis and predictive modeling.
- **Data Science Course – Besant Technologies** (Oct 2024): Focused on EDA, machine learning algorithms, and practical applications using Python, Pandas, and Scikit-learn.

Languages

Telugu • Tamil • English • Kannada

Projects

Hardware Neural Network Accelerator (3-Layer NN with Custom Activation Logic and UART Interface) – M.Tech Project

Designed and implemented a custom RTL-based neural network accelerator without third-party IP cores. The system supports a 3-layer architecture ($4 \rightarrow 8$, $8 \rightarrow 8$, $8 \rightarrow 2$) with 112 on-chip weights (16-bit fixed-point). Implemented four activation functions (ReLU, Sigmoid, Tanh, Linear) using piecewise linear approximations for hardware efficiency. Integrated UART interface (115200 baud) for runtime model swapping and weight updates without re-synthesis. Developed a complete Python host interface for model loading, inference execution, and status monitoring. Achieved inference latency of 280 cycles ($2.8\mu\text{s}$ @ 100MHz) and throughput of 357,000 inferences/second. Served as Project Lead responsible for top-level module design, state machine implementation, system integration, and verification. Validated through comprehensive SystemVerilog testbench with multiple test patterns (positive, negative, zero, large values). All functional requirements met – ready for FPGA/ASIC deployment.

Malware Detection Using Machine Learning Algorithm

I elevated malware detection and classification performance by comparing machine learning algorithms: Gradient Boosting, LSTM, Random Forest, MLP, and RNN. This involved optimizing models, data preprocessing, and seamless integration, resulting in enhanced cybersecurity defenses and real-time threat detection.

Banking Application (Python, Django, Tkinter, PostgreSQL)

Developed a comprehensive banking application using Python, Django for the web interface, and Tkinter for desktop interactions. Implemented core banking functionalities such as account creation, balance checking, deposits, and withdrawals, using a PostgreSQL database for data management. Demonstrated proficiency in handling database connections and executing SQL queries within the Python application.

Telecom Customer Churn Prediction Using Random Forest

Conducted a Telco Customer Churn prediction project, following standard data science procedures: data preprocessing (cleaned the dataset, handled missing values, encoded categorical features); exploratory data analysis (EDA) to understand customer churn patterns; feature engineering to create relevant new features; model selection using Random Forest Classifier; model training and evaluation achieved an accuracy of 0.79 with detailed classification metrics. This project demonstrates ability to apply data science techniques to solve real-world business problems, make data-driven decisions, and build predictive models for customer churn prediction.

Telecom Churn Prediction Using Logistic Regression

Worked on a project that involved predicting telecom customer churn using Logistic Regression. The project consisted of 21 predictor variables, and involved merging and analyzing customer data to forecast whether a customer would switch telecom providers. Demonstrated the complete process of data analysis, modeling, and evaluation for telecom churn prediction.

Sample Super Store Sales Report Using Tableau

Created a comprehensive sales report using Tableau, analyzing key performance metrics for a retail business and highlighted total sales of 1.48 million dollars and a profit margin of 13.09%. Provided insights into region-wise sales and state-level performance. Visualized category-wise profits, subcategory-wise sales, and customer purchasing patterns, identifying top-performing products and sales growth trends. Utilized data correlations between sales, profits, and discount rates to enhance business decision-making. Produced dynamic dashboards showcasing key insights on average order values, order quantities, and sales trends over time.

Skills

Technologies: Python, Excel, Advance Excel, Django, SQL, Power BI, Tableau, Probability, Statistics, Machine Learning, Deep Learning, Natural Language Processing, Artificial Intelligence.

Soft Skills

Communication, Problem-solving, Adaptability, Critical thinking, Active listening.